SUDARSHAN UAV — TROUBLESHOOTING GUIDE

Symptom → Cause → Fix | Issue 1.0

Use this guide to diagnose and resolve faults systematically. Work through the Diagnostic Steps in order before applying the Fix. Est. Time is bench repair time excluding re-flash where noted.

Section 1 — Boot & Connectivity

Symptom	Likely Cause	Diagnostic Steps	Fix	Est. Time
IMU not found on I2C scan	Wrong I2C address or loose wire	1. Check SDA/SCL wiring 2. Scan I2C bus (0x68/0x69) 3. Check AD0 pin state	Re-seat connector; set AD0 LOW for 0x68; replace if dead	5 min
Motor driver (Uno) no response	UART mis-wired or firmware not flashed	1. Check Mega TX1 → Uno RX wiring 2. Open Uno serial monitor 3. Verify 'READY' message	Re-wire UART; reflash Uno firmware; verify baud = 115200	10 min
Wi-Fi AP 'SUDARSHAN_AP' not visible	ESP32 not powered or firmware issue	1. Check 5V supply to ESP32 2. Check TX2/RX wiring 3. Monitor ESP32 serial	Power-cycle ESP32; reflash firmware; verify AP SSID in code	10 min
GCS shows 'Connection Refused'	Wrong IP/port or ESP32 not in AP mode	1. Confirm IP = 192.168.4.1 2. Confirm port = 5760 3. Ping 192.168.4.1	Update GCS config; confirm ESP32 AP mode; check firewall	5 min
ESC no startup beep	No throttle signal or ESC fault	1. Check ESC PWM cable 2. Verify PCA9685 I2C 3. Check 5V to PCA9685	Re-seat PCA9685 wiring; run ESC calibration; replace ESC if silent	15 min

Section 2 — Flight Instability

Symptom	Likely Cause	Diagnostic Steps	Fix	Est. Time
Roll/pitch oscillation during hover	Kp too high in roll/pitch PID	1. Observe frequency of oscillation 2. Check Kp value in firmware 3. Log PID output	Reduce Kp_roll / Kp_pitch by 10–15%; re-tune Kd if needed	20 min
Lateral drift in hover	IMU levelling offset or motor imbalance	1. Place on level surface, check roll/pitch = ~0 2. Run MOTOR_TEST 3. Check prop balance	Calibrate IMU level offsets; balance or replace props; check motor thrust	30 min
Yaw drift with no yaw command	Motor direction / prop mismatch or IMU_YAW_SIGN wrong	1. Verify CW/CCW prop placement 2. Check yaw PID output log 3. Review IMU_YAW_SIGN	Correct prop directions; set IMU_YAW_SIGN=-1 if reversed; retune Ki_yaw	15 min

Symptom	Likely Cause	Diagnostic Steps	Fix	Est. Time
Uncontrolled altitude climb	Altitude PID Kp too high or sonar noise	1. Monitor alt_cm for spikes 2. Check sonar_ok flag 3. Review Kp_alt value	Reduce Kp_alt by 10%; add sonar median filter; check sonar alignment	20 min
Motors running at uneven speeds	ESC not calibrated or motor map wrong	1. Run MOTOR_TEST — note uneven response 2. Check motor map 3. Check ESC calibration	Re-run ESC calibration; verify SET_MOTOR_MAP; replace suspect ESC	25 min

Section 3 — Sonar & Altitude

Symptom	Likely Cause	Diagnostic Steps	Fix	Est. Time
sonar_ok = 0 reported	Sonar wiring fault or out of range	1. Check TRIG/ECHO wires on Mega D9/D10 2. Test with hand at 20 cm 3. Monitor raw echo time	Re-seat sonar connector; clear any obstruction in beam path; replace sensor	10 min
alt_cm jumps suddenly to 0 or 999	Sonar echo timeout / electrical noise	1. Check sonar power (5V) 2. Look for nearby motor EMI 3. Add 100µF cap on sonar power	Add decoupling capacitor; route sonar wires away from motor lines; shield cable	15 min
False landing detected mid-flight	Sonar beam reflected off grass / angled surface	1. Check sonar tilt angle 2. Fly over flat surface 3. Review consecutive-count threshold	Ensure sonar is vertical; increase LAND_DETECT_COUNT in firmware	10 min

Section 4 — GCS & Communication

Symptom	Likely Cause	Diagnostic Steps	Fix	Est. Time
Telemetry display frozen	TCP socket disconnect or FC loop stall	1. Check GCS socket status indicator 2. Ping ESP32 3. Check FC serial for READY	Reconnect GCS; power-cycle ESP32; check FC loop timing	5 min
Commands not reaching FC	UART wiring fault or baud mismatch	1. Monitor ESP32 serial — does it receive? 2. Check Mega RX wiring 3. Verify baud = 115200	Fix ESP32→Mega wiring; verify firmware baud rates match	10 min
Web GCS shows LOCKED	Another session holds priority lock	1. Check if another device is connected 2. Enter override code 1410 3. Wait 60 s for timeout	Use override code 1410; disconnect other session; power-cycle ESP32	2 min
Sequence gap warnings in GCS log	UDP/TCP packet loss or FC overload	1. Check Wi-Fi signal strength 2. Monitor FC loop timing 3. Review GCS log for pattern	Move closer to AP; reduce GCS telemetry rate; check FC CPU load	10 min

Section 5 — Battery & Power

Symptom	Likely Cause	Diagnostic Steps	Fix	Est. Time
bat_mv reads 0 or static	Voltage divider fault or ADC pin float	1. Measure battery pin with multimeter 2. Check ADC resistor divider 3. Check Mega pin A0	Repair voltage divider; check solder joints; verify VBAT_PIN in firmware	15 min
Auto-LAND triggers at full battery	VBAT_LOW threshold too high or divider ratio wrong	1. Compare bat_mv to multimeter reading 2. Check VBAT_SCALE constant 3. Verify resistor values	Recalculate and update VBAT_SCALE; verify R1/R2 divider ratio	20 min
ESCs beep continuously after power-on	No valid PWM signal or ESC in error state	1. Check PCA9685 outputs with oscilloscope 2. Verify ESC power 3. Check Uno READY status	Re-run ESC calibration; reflash Uno; check PCA9685 I2C address (0x40)	20 min

Section 6 — Preflight Test Failures

Symptom	Likely Cause	Diagnostic Steps	Fix	Est. Time
MOTOR_TEST timeout — no motor response	Uno not responding or wrong motor map	1. Check Mega→Uno UART 2. Open Uno serial monitor 3. Verify motor map order	Reflash Uno; fix UART wiring; run SET_MOTOR_MAP; re-run MOTOR_TEST	15 min
IMU preflight fail	IMU not found or calibration offset too large	1. Check I2C scan (expect 0x68) 2. Place drone on level surface 3. Review gyro drift in log	Fix I2C wiring; recalibrate IMU offsets; replace MPU-6050 if unresponsive	20 min
Sonar sanity fail	alt_cm outside 5–200 cm range at bench	1. Check sonar wiring 2. Place hand 30 cm from sensor 3. Verify TRIG/ECHO pin config	Fix sonar wiring; clear beam path; replace sensor if no pulse detected	10 min